Aleksander Nesvik, Daniel Alnes

Visualisation of AIS message for the Norwegian Coastal Administration

A more secure ocean for the norwegian people

Bachelor's thesis in computer engineering
Supervisor: Di Wu
Co-supervisor: None
May 2026

Norwegian University of Science and Technology
Faculty of Information technology and electrical engineering
Department of ICT and natural science

NTNU

ABSTRACT

Write an abstract/summary of your thesis, and state your main findings here.

A summary should be included in both English and any second language, if this is applicable, regardless if the thesis is written in English or in your preferred language. These should be on separate pages, the English version first.

PREFACE

Write the preface of your thesis here.

You may include acknowledgements and thanks as part of your preface on this page, or you may add it as a new chapter after the preface.

CONTENTS

LIST OF FIGURES

LIST OF TABLES

ABBREVIATIONS

List of all abbreviations in alphabetic order:

- **AIS** Automatic Identification System
- **NCA** Norwegian Coastal Administration
- **NTNU** Norwegian University of Science and Technology

TERMS

- **WebGL** Web Graphics Library, is a javascript api that allows GPU-Accelerated rendering of websites in browser, `https://en.wikipedia.org/wiki/WebGL`
- **Vector tiles** Geographic data is passed into tile shapes, which is used to show for example a world map rendered in browser `https://en.wikipedia.org/wiki/Vector_tiles`
- **JSON** lorem shipsum
- **GeoJSON** lorem shipsum
- **Deck.gl** lorem shipsum
- **SuperCluster.js** lorem ipsumm

CHAPTER

ONE

INTRODUCTION

This chapter presents the background of the project, outlines the objective and defines the requirements

1.1 Background

Information about the seafaring vessels around the Norwegian coast are paramount for national security and the security of said vessels, thus it is paramount that such information is available in a concise and easy to understand format for the professionals working in the Norwegian coastal administration. Every day millions of anomaly messages are sent out from a ships AIS messaging unit, and parsing through all these messages to see which are more important than others is a significant and important task for our government.

1.2 Objective

The objective of this project is to develop a web-based program that can visualise AIS anomaly data to provide maritime experts with better usable insights. The program will also serve as a tool for both live monitoring and historical analysis.

Specific goals include:

- **Advanced Map Visualization:** Implementing a map interface using MapLibre to display anomaly clusters, satellite paths, and base station locations.
- **Data Aggregation and Clustering:** Developing methods to handle high-density data without overwhelming the user interface.
- **Temporal Analysis:** Provide a time based feature that allows users to visualise data forward and backward in time to observe development of anomalies.
- **Efficient Querying:** Building a bridge between the React frontend and the backend database to allow users to filter data by ship type, anomaly category, and time intervals.

1.3 Motivation

During a normal day the NCA gets around 80 million AIS messages that are suspected and flagged as anomalies.

1.4 Requirements

The solution will be implemented as a local database and a website that can query the data it needs and visualise that information. All data processing will be handled on the device hosting the database.

1.5 Project description

From the NCA we were tasked with creating a data-visualization software that better displays crucial information about anomalies, than the one they already have in use.

The software in use today has a lack of features that the client wishes to have, like different visualization methods. i.e(heatmapping, hexagonal visualization) etc.

This project is used as a prototype for software that kystverket themselves can take further, where all data used for visualization is dummy data. It was expected that we work in the React programming language, while using libraries and frameworks that they already use at the NCA.

1.5.1 Stakeholders

- **The Norwegian Coastal Administration (Kystverket):** The primary stakeholder and "client." They provide the problem description, the tech stack requirements (Azure, .NET, React), and the AIS data.
- **The Project Team:** Two students from NTNU responsible for the design, implementation, and testing of the visualization prototype.
- **Maritime Operators:** The end-users who require an intuitive interface to maintain maritime safety.

1.6 Thesis structure

The report will be used to outline how the problem was solved based on the problem description in section 1.2, how this was solved technically, what tools were used etc.

- **Chapter 2 – Theory:** will be delving into the theory for the project, what if any equations the software was based on. This will give the underlying theory on how mapping visualization works, How previous research in visualization of in browser map systems, and fast clustering and or showcasing of data.

- **Chapter 3 – Methods:** focuses on the methodology and hardware the group used during the project, what libraries, programming languages and libraries, also how the project was structured from start to finish.

- **Chapter 4 – Results:** describes the results the group accomplished, how it was accomplished, why certain solutions to problems were used, describing the process and solution in detail.

- **Chapter 5 – Discussion:** discusses the project in detail. It explains the group members' thoughts on the project, how the plan was followed, if there were discrepancies between the plan and the actual progression of the project.

- **Chapter 6 – Conclusion:** serves as a conclusion to the report and discussion on future development on the software that can be done, our recommendations

- **Chapter 7 – Societal Impact:**

CHAPTER

TWO

THEORY

2.1 Equations

A simple equation can be included as:

$$r = 2\pi^2 \tag{2.1}$$

The text below shows an example of how to align equations on the equal sign, with only one reference for both. This may be useful for when they are linked and are actually only one equation but splitting them up makes it more readable.

$$\begin{aligned} a &= \sin^2(\Delta\phi/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2(\Delta\lambda/2) \\ d &= 2R \cdot \arcsin(\sqrt{a}) \end{aligned} \tag{2.2}$$

The whole equation can be referenced as "equation (2.2)", here showing the Haversine formula. One may also align sub-equations such that they are numbered the same but have a letter differentiating them as shown below. This can be used when they are linked, but you will need to reference both individual parts.

$$SSD = \sum_{i=1}^{n} (\vec{x_i} - \vec{\mu_q})^2 \tag{2.3a}$$

$$SSE = \sum_{q=1}^{k} \delta_{rq} SSD \tag{2.3b}$$

These equations can be refrenced by their spesific sub-equation as "equation (2.3a)", or by the whole group as "equations (2.3)". The "double backslashes" in the .tex creates line spaces and gives more room around the equations and paragraphs. Use them as you think feels right.

2.2 Tables and footnotes

Here is an example of both a regular table with data and a table with split headers, for scientific usage. Do not use horizontal/vertical rulers between the data, or encase the table with rulers.

Statistic	**Velocity**	**Altitude**	**1/Angle**	**Temp.**
Mean	122.68	240.98	93.75	13.95
Std	224.51	145.88	60.39	4.44
Q1	28.00	111.60	34.15	10.60
Median	63.00	223.20	99.59	13.30
Q3	137.00	359.10	151.99	16.70
Min	0.00	1.00	0.00	3.30
Max	14519.00	616.70	180.00	32.10

Table 2.2.1: Table of dynamic feature statistics where outliers are included, for all data points. Velocity is given in m/h, the altitude in *mamsl*, the inverse trajectory angle in 1/degrees, and temperature in degrees Celsius.

Area 1	**Start date**	**End date**
2018	03.06	29.06
2019	03.06	03.07 or 31.08[1]
2020	03.06	05.09
Area 2	**Start date (farm 1/2)**	**End date**
2012	09.06	07.09
2013	23.06 / 15.06	25.08
2014	05.06 / 25.06	10.09
2015	13.06 / 03.07	06.09
2016	17.06	22.07

Table 2.2.2: Selected time ranges for the data in all areas and all years.

[1] A footnote explaining something.

2.3 A single figure

Figure 2.3.1 is included as an example. The square brackets before the caption description contains the title of the figure, which is what will be written in the list of figures. This should be short and concise. The same layout applies to tables and other floats. Remember to change the title as well as the caption if you are copying these examples. In the list of figures and tables all the different floats will be grouped together by chapter. Remember to always reference all your figures and tables in the text at least once.

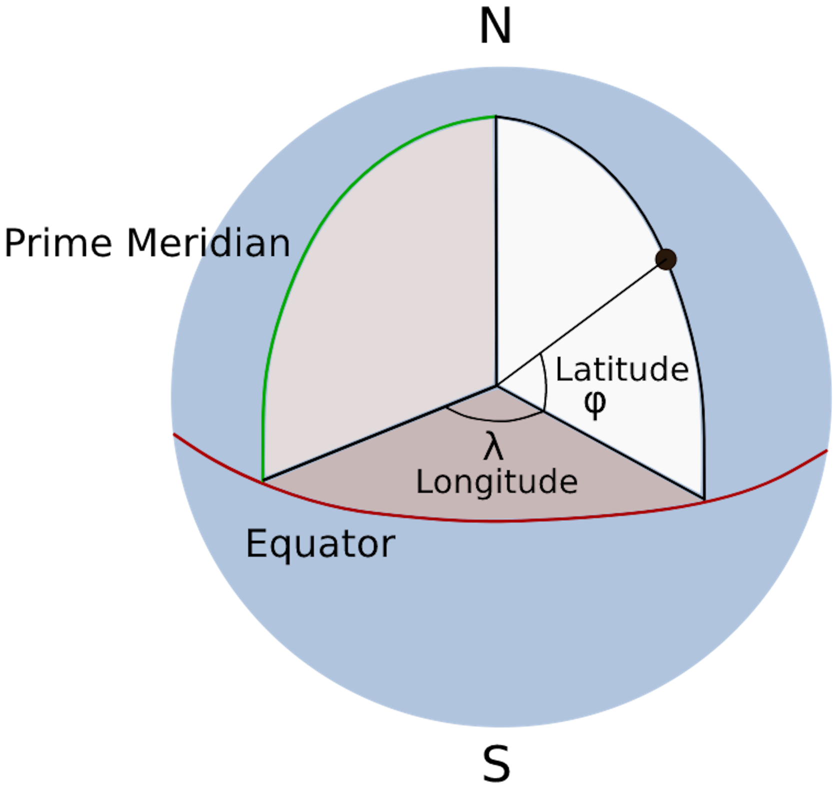


Figure 2.3.1: Illustration of the earth, and how latitudes and longitudes are calculated with respect to the equator and the prime meridian.

2.4 Citations

Here are some examples on how to reference a source (where none is relevant to the text but just for illustration purposes only). One may cite a single reference by calling [**wolves_of_mount_mckinley**], or several in the same bracket by calling [**machine_learning**, **clustering_impossibility**] when they are all related to the same statement. There are many different styles on how to cite, and how the layout and order of your citation style is presented. This is my favorite, as I find it neat and tidy [**sheep**]. It will show up in order of appearance in the references section.

CHAPTER
THREE

METHODS

Include the complete description of the methods used in your research here.

Below is an example of how subsectioning works. The sections and subsections will be included in the table of contents, while subsubsections will not be in the table of contents but still have their own title in the text.

3.1 Section one

3.1.1 Subsection one

3.1.1.1 Subsubsection one

3.1.1.2 Subsubsection Two

3.1.2 Subsection Two

3.2 Section two

CHAPTER

FOUR

RESULTS

4.1 More figures

This section includes some examples of different types of figures to include. A simple single figure is shown in figure 4.1.1, while figure 4.1.2 shows how three subfigures can be included together. Remember to change both the caption and the title in the square brackets before the caption, which will show up in the list of figures.

page fill page fill.

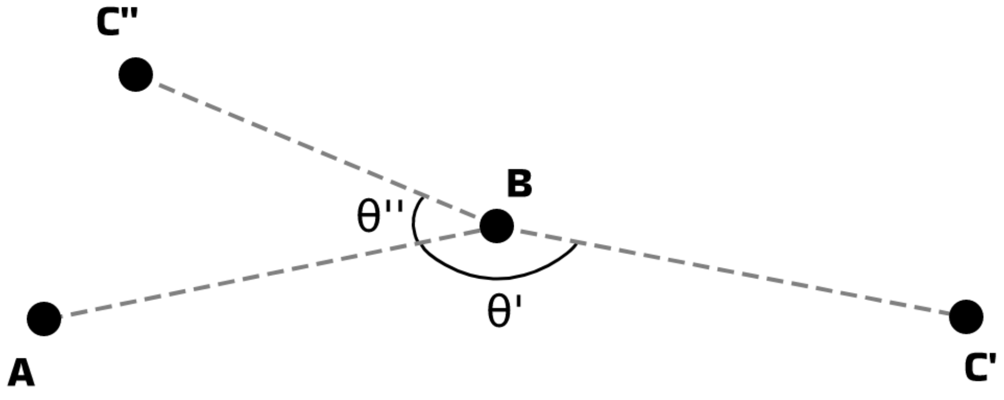
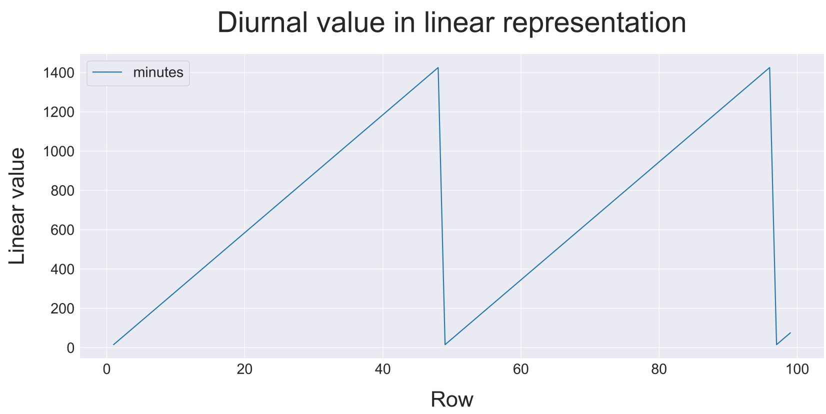
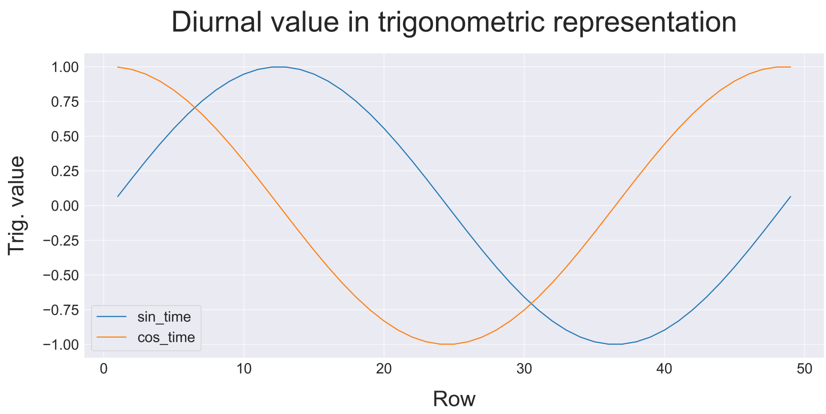


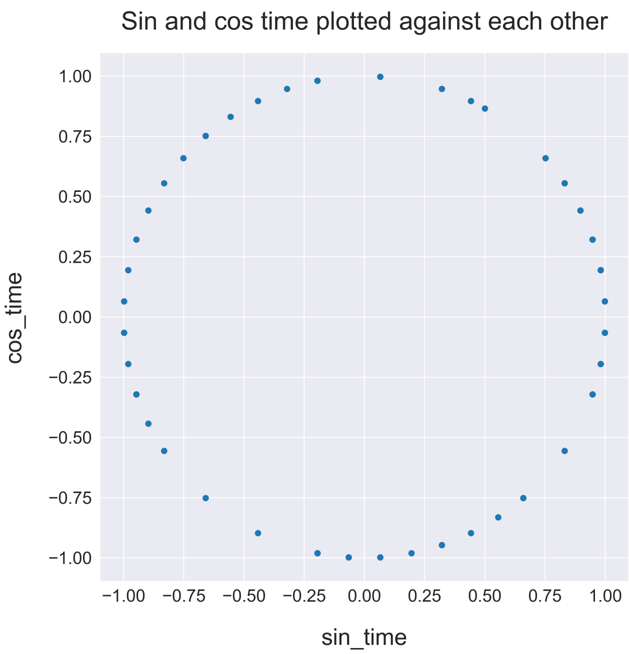
Figure 4.1.1: The trajectory angle found for the trajectory ABC between the three points A, B and C. The two different C-points show the angle gotten for relatively unchanged directional trajectory with C', and opposite directional trajectory with C''.



(a) Time as a linear variable.



(b) Time represented as pairs of sine and cosine.



(c) Time as a cyclic feature.

Figure 4.1.2: Figure (a) shows the time variable in the data for the first 100 rows using linear time as minutes past midnight, figure (b) shows the trigonometric time representation for one cycle where each time point has a unique sine and cosine-pair value, and figure (c) shows time as a cyclic feature with the trigonometric pairs.

CHAPTER

FIVE

DISCUSSION

Discuss your results here.

5.1 Future work

Include a section about what should or could be done in future research, or explain any recommended next steps based on the results you got. This should be the last section in the discussion.

CHAPTER

SIX

CONCLUSIONS

Give a concise summary of your research and finding here, and include a short summary of any future work as well.

APPENDICES

A - GITHUB REPOSITORY

All code and latex-files used in this document are included in the Github repository linked below. Further explanations are given in the readme-file.

Github repository link

- `https://github.com/ninasalvesen/thesis_latex_template`

B - SIDENOTE STATISTICS

B1 - Some random table

Remember to only include one thing per page in the appendices.

Statistic	**One**	**Two**
Count	387317	283960
Mean	130.66	134.18
Std	248.09	230.32
Q1	31.00	21.00
Median	67.00	63.00
Q3	142.00	159.00
Min	0.00	0.00
Max	14519.00	14253.00

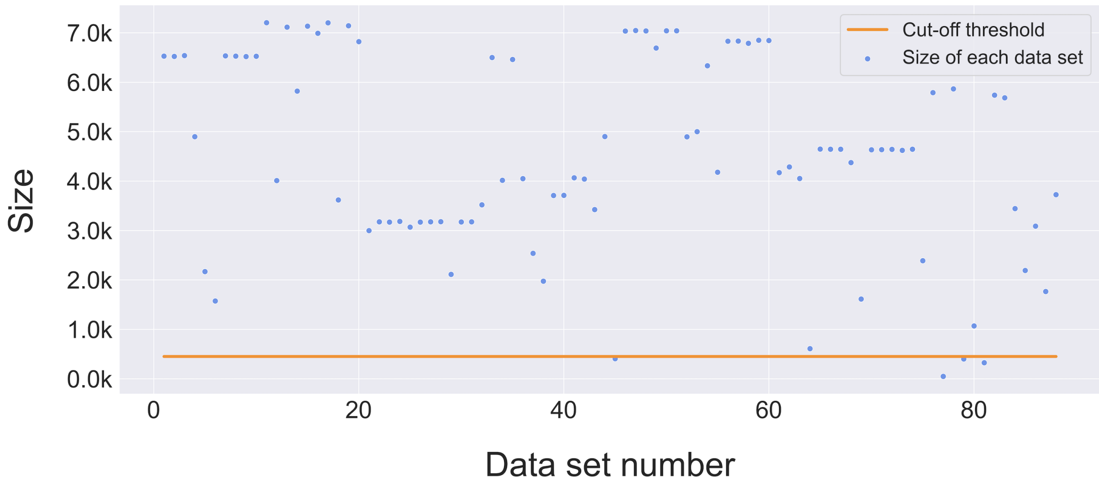
Table B.1: Table of statistics on some sidenote data.

B2 - Some other random table

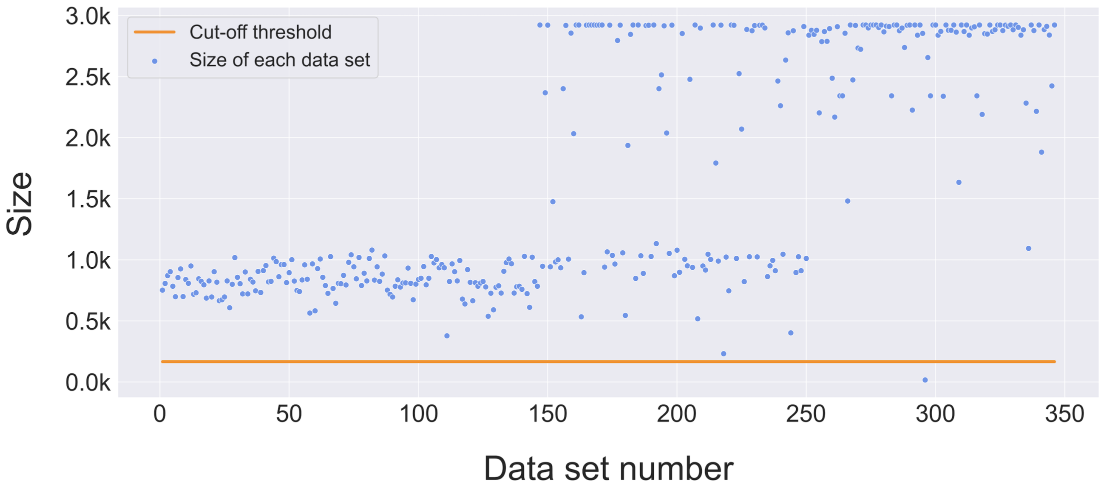
Statistic	**Three**	**Four**
Count	387317	283960
Mean	130.66	134.18
Std	248.09	230.32
Q1	31.00	21.00
Median	67.00	63.00
Q3	142.00	159.00
Min	0.00	0.00
Max	14519.00	14253.00

Table B.2: Table of statistics on some other sidenote data.

B3 - Some random figure



(a) Data set sizes for data X.



(b) Data set sizes for data Y.

Figure B.1: The figures show the data set sizes of X and Y and the proposed cut-off threshold at 10 % of the mean set size.